PART 7

Deployment & Release

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#051

Feature Flags

WHAT IT IS

Toggles that turn features on/off at runtime without deploying.

WHY IT MATTERS

Ship dark, roll out gradually, instant kill-switch, A/B tests.

EXAMPLE

```
Feature::define('new-board', fn (User $u) =>
    $u->isBetaTester() || lottery(odds: 10, outOf:
    100));

if (Feature::active('new-board')) { return view(
    'board-v2'); }
```

#052

Blue-Green Deployment

WHAT IT IS

Two identical environments: Blue (live) and Green (new). Switch traffic after Green passes checks.

WHY IT MATTERS

Near-zero downtime and instant rollback by switching back.

EXAMPLE

Deploy v2 to Green → run smoke tests → flip load balancer/DNS to Green → keep Blue warm for rollback. DB migrations must be backward compatible with both versions.

#053

Canary Releases

WHAT IT IS

Send a small percentage of traffic to the new version first and watch metrics.

WHY IT MATTERS

A bad release only hurts 1-5% of users.

EXAMPLE

```
upstream app {
    server v1.internal:8080 weight=95;
    server v2.internal:8080 weight=5;
}
```

#054

Rolling Deployments

WHAT IT IS

Replace instances gradually (a few at a time) with the new version.

WHY IT MATTERS

No downtime and no need for double infrastructure.

EXAMPLE

```
strategy:
  type: RollingUpdate
  rollingUpdate: { maxUnavailable: 0, maxSurge: 1 }
```

#055

Rollbacks

WHAT IT IS

Returning to the previous known-good version.

WHY IT MATTERS

The fastest mitigation for a bad release is usually undoing it.

EXAMPLE

```
kubectl rollout undo deployment/taskflow-web
```

#056

Health Checks

WHAT IT IS

An endpoint or command that reports whether the app is working.

WHY IT MATTERS

Load balancers and orchestrators use it to route traffic and restart failures.

EXAMPLE

```
Route::get('/healthz', function () {
    DB::select('select 1');
    Redis::ping();
    return response()->json(['status' => 'ok']);
});
```

#057

Liveness & Readiness Probes

WHAT IT IS

Kubernetes checks: liveness = "is the process stuck? restart it"; readiness = "can it take traffic right now?".

WHY IT MATTERS

Prevents routing traffic to booting/overloaded pods and restarts hung ones.

EXAMPLE

```
livenessProbe:
  httpGet: { path: /up, port: 8080 }
  periodSeconds: 10
  failureThreshold: 3
readinessProbe:
  httpGet: { path: /healthz, port: 8080 }   # checks
    DB/Redis
  periodSeconds: 5
```

Found this useful?

Save it for later.

Share it with a teammate.

Full guide with all code examples:
the Backend Engineering Handbook (GitHub).